

Shruti Jayaraman

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EDUCATION

University of Michigan

Master of Science, Computer Science and Engineering | GPA: 3.7/4.0

Ann Arbor, MI

Aug 2025 – May 2027 (Expected)

Coursework: NLP, Randomness and Computation, Scalable Systems for Agentic AI, Computer Vision, Compilers, Reinforcement Learning

College of Engineering, Guindy (Anna University)

Bachelor of Engineering, Computer Science and Engineering | GPA: 9.3/10

Chennai, India

Jul 2019 – Apr 2023

TECHNICAL SKILLS

Languages: Python, C++, Java, C, SQL, TypeScript, JavaScript, Bash

ML/AI: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, LangChain, LangGraph, RAG, LLM Evaluation, Stable Diffusion

Data & MLOps: pandas, NumPy, Weights & Biases, Docker, Git, GitHub Actions (CI/CD), FastAPI, Streamlit, Linux, SLURM, Amazon Web Services

Databases: MySQL, SQLite, ChromaDB

Domains: Machine Learning, NLP, LLM Agents, Multi-Agent Systems, RAG, LLM Evaluation, CI/CD, Code Review, Data Pipelines

WORK EXPERIENCE

Caterpillar Inc.

Automation Engineer

Chennai, India

Jan 2023 – May 2025

- Validated a GNSS/IMU sensor fusion system for autonomous mining vehicles operating in high-altitude and underground environments, achieving **15% latency reduction** in real-time navigation and improving positional accuracy in GPS-degraded conditions
- Extended a proprietary simulation platform with **2 physics engines** (Bullet3, ODE) to model complex underground terrain dynamics, replacing physical test-site deployments in the validation cycle
- Engineered a geospatial ETL pipeline in Python producing interactive map visualizations for mine site planning
- Automated discrepancy flagging across **4 databases (12,000+ records each)**, cutting manual validation effort by **60%**

RESEARCH EXPERIENCE

Blablablab, University of Michigan

Student Researcher (advised by Prof. David Jurgens)

Ann Arbor, MI

Jun 2026 – Present

- Implemented a Python pipeline classifying **5,250 LLM-generated candidate personas** against an 11-category bias codebook across **5 model families** on the Anthropic API
- Surfaced a stealth bias loophole and judge-generator hypocrisy in a 2-judge evaluation harness

LIT Lab, University of Michigan

Student Researcher (advised by Prof. Rada Mihalcea)

Ann Arbor, MI

Oct 2025 – Present

- Lead an independent study on which components of agent skill files drive agent execution fidelity; formulated a span-level attribution methodology testing **7 skill-file features** across **3 deterministic agentic tasks** with open-source models
- Co-designed an ACL-submitted clinical NLP system for medical visit preparation and its RAG-based evaluation framework across **20 human and 90 simulated patients**, decreasing patient mental effort by **19%** ($4.80 \rightarrow 3.90$, $p = 0.014$) vs. a plain LLM baseline

PROJECTS

MemHub: A Memory-as-a-Service Solution for Agentic AI

Apr 2026

- Built a 2-tier SQLite + ChromaDB memory layer with confidence-gated retrieval for multi-agent LLM systems, resolving context-window overflow across agents
- Lowered latency **6%** and boosted memory-budget adherence **25%** vs. a single-tier baseline across **7 benchmarks**

RepoRanger: Privacy-first AI Code Reviewer & Automated Branch Janitor

Apr 2026

- Launched a Dispatcher-Worker GitHub App with AI-powered PR reviews (Groq/Llama-3.3-70b) and keyword-triggered automated branch hygiene, earning **44 GitHub stars**
- Architected the app to run entirely on user infrastructure via GitHub Actions at **\$0 infrastructure cost**, minimizing reliance on paid code review tools

Git-Hired: AI Powered Adaptive Interview Platform

Nov 2025

- Shipped a 4-agent AI interview platform (FastAPI, React/TypeScript, WebSockets) where a 5-tier scoring loop adapts question difficulty **± 2 levels** mid-interview
- Developed a syntax-agnostic evaluator that grades code logic and auto-delivers **100-point reports** to hiring managers over SMTP

PUBLICATIONS

- Jayaraman, S. et al. "TexAVi: Generating Stereoscopic VR Video Clips From Text Descriptions." *3rd IEEE International Conference on Computer Vision and Machine Intelligence (CVMI)* 2024.